R&D Projects and Publications

R&D Projects (2013 - 2020)

An artificial Intelligence based robot for sorting plastic bottles:
It is a joint work with a trash recycling company. The objective is to detect, classify and separate plastic bottles from other garbage placed on a moving conveyer belt. For detection and classification, we used a hyper-spectral camera and for bottle collection, we used an Ethercat based manipulator. I worked on the development of image processing and machine learning algorithms. Also, I worked on low level system integration and designed the whole system architecture.

A mobile platform for SLAM with multiple cameras, an IMU, and a GPS:
It is a joint work with a Researcher Center. The main system has been developed using ROS with C++ and Python. I worked as the system architect. I developed the platform software. Also, I contributed to the implementation of SLAM algorithms.

An Industry 4.0 based project:
It is a joint work with a large manufacturing company. We worked on *"Big Data"* based predictive maintenance of various parts in the main production line.

TRUST - a humanoid robot:
It is a humanoid robot with 35 major joints capable of doing biped walking, stair climbing, and simple manipulation. We used both Ethercat and CAN communication and Elmo motor controllers to control most of the joints. For some of the joints, we used our own motor controllers. The overall motion system has been developed using C, C++ on Xenomai. I designed the whole software system architecture and implemented software platform. I also contributed to the controller design and implementations.

Tibo - a Ski-robot:
We participated in "Winter Olympics 2018: robot ski tournament" with Tibo. I designed the overall system architecture and implemented the platform. Also I contributed to trajectory generation algorithms.

RailBot:
It is a small rail robot for remote monitoring: This robot has an image camera and a thermal camera. It moves autonomously and transfers images and other information to some remote user. I worked on the development of the motor controller to control the speed of the robot. I also developed the mobile app for android devices.

A temperature monitoring system:
It is a temperature monitoring system of cooling water for industrial use. We developed it for some manufacturing company. I developed the whole firmware on Microcontroller - STM32.

Dissertations

Mohammad Khairul Hasan *Approximation Algorithms for Facility Location and Graph Partitioning Problems.* (Ph.D.). Korea Advanced Institute of Science and Technology, 2013

Mohammad Khairul Hasan *Improved approximation algorithms for connected facility location problems.* (MS). Korea Advanced Institute of Science and Technology, 2006

Journal Publications

Mohammad Khairul Hasan, and Kyung-Yong Chwa. *Approximation algorithms for the Weighted t-Uniform Sparsest Cut and some other graph partitioning problems.*J. Comput. Syst. Sci. 82(6): 1044-1063, 2016.

Md. Shamim Ahsan, Man Seop Lee, Mohammad Khairul Hasan, Young-Chul Noh, Farid Ahmed, Martin B.G. Jun. *Formation mechanism of self-organized nano-ripples on quartz surface using femtosecond laser pulses* .Optik - International Journal for Light and Electron, 126(24), 2015.

Kunho Kim, Mohammad K. Hasan, Jae-Pil Heo, Yu-Wing Tai, Sung-eui Yoon. *Probabilistic cost model for nearest neighbor search in image retrieval* Computer Vision and Image Understanding, 116(9): 991-998, 2012.

Hyunwoo Jung, Mohammad Khairul Hasan, and Kyung-Yong Chwa. *A 6.5 factor primal-dual approximation algorithm for the connected facility location problem.*J. Comb. Optim., 18(3):258-271, 2009.

Mohammad Khairul Hasan, Hyunwoo Jung, and Kyung-Yong Chwa. *Approximation algorithms for connected facility location problems.* J. Comb. Optim., 16(2):155-172, 2008.

Conference Publications

Mohammad Khairul Hasan, Sung-Eui Yoon, and Kyung-Yong Chwa. *Bounds on the geometric mean of arc lengths for bounded-degree planar graphs.* In FAW, pages 153-162, 2009.

Hyunwoo Jung, Mohammad Khairul Hasan, and Kyung-Yong Chwa. *Improved primal-dual approximation algorithm for the connected facility location problem.* In COCOA, pages 265-277, 2008.

Mohammad Khairul Hasan, Hyunwoo Jung, and Kyung-Yong Chwa. *Improved approximation algorithm for connected facility location problems.* In COCOA, pages 311-322, 2007.